

ACADEMIC YEAR 2026/2027	 Prime School of Portugal	YEAR 6 LOWER SECONDARY
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SUBJECT: ICT

YEAR 6

ACADEMIC YEAR: 2026-2027

Robotics & ICT

Teacher: Sara Jorge

GENERAL GOALS	SPECIFIC GOALS WITH ROBOTIC PLATFORMS & ACTIVITIES & CODING LANGUAGE	CONTENT
The assessment objectives (AOs) The curriculum provides students with opportunities to develop computational thinking, problem-solving, and digital literacy skills through engaging, hands-on STEM activities. Learners are encouraged to explore how computer-controlled systems function and to apply their knowledge in practical contexts by designing, programming, and evaluating technological solutions. Students follow the engineering design process to investigate challenges, develop and test prototypes, analyses outcomes, and refine their designs. This process promotes creativity, resilience, critical thinking, and collaborative learning while encouraging students to reflect on and improve their work. The programme introduces the fundamental principles of robotics by combining basic robot construction with programming through coding. Students develop an understanding of algorithms, sequencing, logical reasoning, and the relationship between hardware and software in controlling robotic systems.	Chapter 1: Our Digital World - Develop an understanding of digital citizenship and demonstrate safe, responsible and respectful online behaviour. Protect personal information and maintain a positive digital identity and digital footprint. Apply online safety practices, including creating strong passwords and recognising online risks. Evaluate the reliability of digital information using critical thinking skills. Create digital content that promotes responsible and ethical use of technology. Chapter 2: Content Creation - Plan and create effective digital presentations for different audiences and purposes. Use text, images and multimedia to communicate information clearly and creatively. Apply basic design principles to improve the quality and appearance of digital content. Respect copyright by using and acknowledging digital resources appropriately. Evaluate and improve digital work through reflection and feedback.	1st term Chapter 1: Our Digital World • Digital citizenship and online responsibility. • Digital identity and online reputation. • Privacy and personal data. • Cybersecurity and protecting digital accounts. • Misinformation, disinformation and fake news. • Evaluating online sources. • Ethical use of technology. • Artificial Intelligence and its impact on society. Chapter 2: Content Creation • Planning and designing digital media. • Advanced presentations and multimedia. • Image, audio and video editing. • Copyright and Creative Commons licences. • Creating digital products for different audiences.

Learners also explore the fundamentals of electronics by creating and testing simple circuits, developing practical technical skills and an understanding of electrical components and their functions. Throughout the course, students are encouraged to communicate their ideas effectively, work collaboratively, and apply safe and responsible practices when using digital technologies. Overall, the curriculum supports the development of confident, responsible, reflective, innovative, and engaged learners, in line with the Cambridge Learner Attributes, while preparing students for further study in Computing and other STEM disciplines.	Chapter 3: Create with Code - Develop computational thinking by designing and creating block-based programs. Apply sequences, events, loops, conditions and variables to solve programming challenges. Test, debug and improve programs to achieve the desired outcome. Create interactive projects that respond to user input using animations, sounds or physical devices. Demonstrate creativity, logical reasoning and problem-solving skills through coding activities. Chapter 4: How Computers Work - Understand how hardware and software work together in a computer system. Identify and explain the functions of input, processing and output devices. Explore the use of sensors and controllers in automated systems. Understand how computers process and transfer information to perform tasks. Design and evaluate simple computing systems using logical thinking and problem-solving skills. Chapter 5: Connect the World - Understand how computers communicate through networks and the Internet. Explain the roles of clients, servers, routers and data packets in digital communication. Explore how information is transmitted safely and efficiently across networks. Develop computational thinking by creating network simulations using block-based programming. Demonstrate responsible and safe use of networked technologies and online communication. Chapter 6: The Power of Data - Collect, organize and represent data using tables, charts and graphs. Analyze and interpret data to identify patterns, trends and	• Evaluating and improving digital content. • Accessibility and inclusive design. Chapter 3: Create with Code • Designing algorithms. • Variables and data. • Selection (if/else). • Repetition (loops). • Nested programming structures. • Debugging complex programs. • Creating interactive games and simulations. • Physical computing (where available). Chapter 4: How Computers Work • Hardware and software interaction. • CPU, memory and storage. • Binary representation (introduction). • Input, output and processing. • Sensors and control systems. • Embedded systems. • Understanding how computers execute instructions. Chapter 5: Connect the World • Computer networks. • Internet infrastructure. • Data packets. • Routers and switches. • Servers and cloud computing. • Web services. • Network security. • Encryption (introduction). Chapter 6: The Power of Data • Collecting and validating data. • Databases. • Sorting and filtering information.
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	draw conclusions. Use digital tools and sensors to gather and process information accurately. Apply critical thinking to make informed decisions based on data. Present data clearly and effectively to communicate findings and support real-world problem-solving.	• Data visualisation. • Interpreting datasets. • Using spreadsheets. • Data modelling. • Artificial Intelligence and data. • Making evidence-based decisions. Chapter 7: Microsoft Office • Word • Power Point
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Term Teaching Plan

Term 1 4 September – 17 December	Term2 5 January – 19 March	Term 3 5 April – 17 June
Chapter 1: Our Digital World	Chapter 4: How Computers Work	Chapter 6: The Power of Data
Chapter 2: Content Creation	Chapter 5: Connect the World	Chapter 7: Microsoft Office
Chapter 3: Create with Code		

Assessment

Term 1 4 September – 17 December	Term 2 5 January – 19 March	Term 3 5 April – 17 June
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October - Homework in Classroom	January - Homework Microbit classroom	April - Practical Quiz
November - Project 1	February - Project 1	May - Typing test
December - Project 2	March - Typing test	June - Project 1

Assessment Criteria

Term 1 4 September – 17 December	Term 2 5 January – 19 March	Term 3 5 April – 17 June
20% Participation & Effort	20% Participation & Effort	20% Participation & Effort
5% Behaviour	5% Behaviour	5% Behaviour
15% Homework	15% Homework	15% Homework
30% Test	30% Project 1	30% Practical Quiz
30% Project 1	30% Project 2	30% Project

Expected Number of 45-Minute Classes		
1 st Term	2 nd Term	3 rd Term